

Schedule of Water-Cooled Package Water Chillers

Description: WCC-CP-BF-01 to 03

Description	Specified	Offered
Numbers of units	3	3
Type	Fresh Water-cooled, Centrifugal Chiller	Water-Cooled Centrifugal Chiller
Manufacturer	-	YORK
Country of origin	-	PRC
Model Number	-	YKIBKRH95CVG
Cooling Capacity, (kW)	3517	3517
Refrigerant used	HFC	R-134a
Total Input Power, (kW)	-	610.2
COP	5.7	5.76
NPLV (according to AHRI 550/590)	≥ 9.0	10
(1) COMPRESSOR & MOTOR		
Type of compressor	Scroll / Screw / Centrifugal	Centrifugal
Number of compressor	-	1
Compressor speed, (rpm)	<12000-	< 12000
Capacity control type	Inlet Guide Vanne/ VSD	IGV + VSD
Minimum Part Load (%)	-	15%
Motor rating (kW)	-	589 (Overall)
Type of compressor motor starter	Variable Speed Drive	VSD
Full-Load Current, (A)	-	1024
Starting current, (A)	-	0 to Full-load
Type of compressor motor	-	Open Type
Overall equipment power factor	0.85	>0.9
Type of drive	-	Gear

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(2) EVAPORATOR		
Type	Shell and Tube	Shell & Tube
Tube material	Copper	Copper
Tube sheet material	steel-	Steel
No. of passes	-	2
Chilled water entering temp, (°C)	12.5	12.5
Chilled water leaving temp, (°C)	7	7
Chilled water flow rate (L/s)	-	152.5
Working pressure (kPa)	-	1600
Water pressure drop (kPa)	-	72.5
Fouling factor ($m^2 \text{ } ^\circ\text{C/kW}$)	0.018	0.018
Insulating material	-	Closed Cell Foam
Thickness of insulation (mm)	- -	38mm
Water box	Required	Marine Water Box

(3) CONDENSER		
Type	Shell and Tube	Shell & Tube
Heat rejection capacity (kW)	-	4099
Tube material	Copper	Copper
Tube sheet material	Steel	Steel
No. of passes	-	2
Entering water temperature (°C).	32	32
Leaving water Temperature (°C).	37	37
Water flow rate (L/s)	-	196.2
Working pressure (kPa)	-	1600
Water pressure drop (kPa)	-	93.6
Fouling factor ($m^2 \text{ } ^\circ\text{C/kW}$)	0.044	0.044
Water box	Required	Marine Water Box